

Week 8 Lab session

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Section 1. proportion of G/G in a population

Downloaded a CSV file from Ensembl https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39810234-39980235;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel

Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..	Male..	Female..	Unknown..	Genotype..	forward..	strand..	Population..	s..	Father
1				NA19648 (F)				A A	ALL, AMR, MXL	-
2				NA19649 (M)				G G	ALL, AMR, MXL	-
3				NA19651 (F)				A A	ALL, AMR, MXL	-
4				NA19652 (M)				G G	ALL, AMR, MXL	-
5				NA19654 (F)				G G	ALL, AMR, MXL	-
6				NA19655 (M)				A G	ALL, AMR, MXL	-
	Mother									
1		-								
2		-								
3		-								
4		-								
5		-								
6		-								

```
table(mxl$Genotype..forward.strand.)
```

A A	A G	G A	G G
22	21	12	9

```
table(mx1$Genotype..forward.strand.)/nrow(mx1) *100
```

```
      A|A      A|G      G|A      G|G
34.3750 32.8125 18.7500 14.0625
```

Now let's look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(gbr)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1		HG00096 (M)	A A ALL, EUR, GBR	-
2		HG00097 (F)	G A ALL, EUR, GBR	-
3		HG00099 (F)	G G ALL, EUR, GBR	-
4		HG00100 (F)	A A ALL, EUR, GBR	-
5		HG00101 (M)	A A ALL, EUR, GBR	-
6		HG00102 (F)	A A ALL, EUR, GBR	-
	Mother			
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

Find proportion of G|G

```
table(gbr$Genotype..forward.strand.) /nrow(gbr)*100
```

```
      A|A      A|G      G|A      G|G
25.27473 18.68132 26.37363 29.67033
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Lets dig into this further.

Section 4: Population Scale Analysis

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
library(ggplot2)
```

Lets make a boxplot

```
ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```

